

# Yawar Badri

Senior Systems Engineer & Hardware Architect

[linkedin.com/in/yawar-badri](https://www.linkedin.com/in/yawar-badri) | [github.com/wayri](https://github.com/wayri) | Discord: sakha.lin

## Professional Summary

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Full-stack hardware architect specializing in complex systems, hardware pipelines, and emerging energy tech. Proven ability to architect, simulate, and deploy end-to-end solutions spanning from high-power analog frontends and complex PCB layouts to deeply embedded firmware and deterministic desktop host interfaces. Expert in transforming theoretical power concepts into highly reliable physical hardware.

## Core Competencies

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- **Hardware & EDA:** KiCad, Altium, Cadence, SolidWorks, SolidEdge, PCB Layout, Bed of Nails Design
- **Simulation & Modeling:** MATLAB, Simulink, LTspice, Qspice, Root Cause Analysis (RCA), FMECA, WCCA
- **Languages & Firmware:** C, C++, C#, Python, JavaScript, STM32, ARM Cortex, Embedded DSP
- **Test & Measurement:** DSO, MSO, Logic Analyzer, LCR, Megger, PQA, DMM, Hardware-in-the-Loop (HIL)

## Experience

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### Pixxel Space

2022 – Present

Senior Electrical Engineer

Space / High-Rel Electronics

- Leading the end-to-end electronic subsystem design for multiple satellites, specializing in high-reliability (high-rel) electronics engineered to operate in hazardous orbital environments.
- Architecting full Electrical Power System (EPS) end to end, designing power regulation, distribution tiles, and high power density converters.
- Developing Electrical Ground Support Equipment (EGSE), power loggers, input protection devices, and constant current solar cell simulators for rigorous subsystem verification.
- Designed and built the hardware for an in-house GNSS simulator, contributing to vertical integration goals.

### Kashmir University

Technical Advisor

Hardware Startups & Prototyping

- Providing strategic technical architecture guidance to emerging hardware startups.
- Mentoring engineering founders on prototyping-to-production pipelines, component supply chain constraints, and embedded systems integration.

### NIT Srinagar

Project Associate (R&D Architecture)

Power Systems

- Directed hardware architecture and systems integration for an experimental grid-tied inverter testbed.
- Engineered the hardware-software ecosystem required for researchers to deploy and benchmark novel power control algorithms safely, bridging high-level MATLAB models with bare-metal PWM execution.

### University of Kashmir (Sponsored by J&K Govt.)

Research Analyst / Associate

EV Infrastructure Analysis

- Analyzed electrical distribution constraints for emerging EV infrastructure, synthesizing deep technical findings into actionable state infrastructural policy.

## Hardware Architecture & Electronics

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- **10-Channel Power Logger:** Bidirectional DC power logger (100Vmax) featuring custom analog front-end hardware and STM32 USB 2.0 interface. Included hardware design, firmware, calibration, and desktop app.
- **Power Conversion Suite:** Built a 100W linear supply, 100W 2-quadrant supply, and 150W electronic load. Worked on kilowatt-scale grid-integrated PV inverters.
- **DACard-Basic HIL Interface:** Architected a full-stack hardware-in-the-loop testing ecosystem encompassing physical hardware, embedded firmware, host software, and custom Python libraries.
- **Ground Energy Storage BMS:** Designed and deployed comprehensive battery management systems (BMS) for large-scale energy storage solutions.
- **3-Phase Grid-Tied Inverter:** Hardware design layout for 3-phase grid-tied power conversion, engineered to support active and reactive power injection algorithms.
- **DFIG Test Bench Hardware:** Architected a full Doubly Fed Induction Generator test bench suite through the full PCB layout and design phase.
- **Dual-channel Oscilloscope:** Ported and modified code for an ATmega2560 featuring a TFT touch interface, engineered specifically for teaching ADC interrupts.

- **8-channel Optocoupler Board:** Industrial isolation interface utilizing an EL817 array with onboard inverted channel logic specifically meant for control isolation.

## Software & Embedded Tools

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- **CATS (CAN Automated Testing Software):** Used to test hardware with a full command, read, verify, record cycle based on PCAN hardware as an interface.
- **KiWay & Transformer Tools:** PCB Design routing framework for automating redundant constraints, and internal CAD tools to calculate and parameterize custom magnetic transformers.
- **sigLib:** C++ DSP library tailored for real-time waveform conditioning, smoothing, and analysis on embedded targets.
- **Gauss-Seidel Load Flow:** Console-based numerical analyzer for calculating power distribution flows across N-bus non-linear electrical networks.
- **Custom DAQ & Desktop Apps:** Developed bespoke desktop applications tailored for determining electrical load profiles, automated billing, and general data acquisition systems.

## Interactive Engineering Utilities (Web-Based)

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- **Power & Systems Engineering:** Developed interactive tools for **SMPS Topology & Ratio** solving, **Loop Compensators (P/Z)**, **AC Motor Dynamics**, **PV Yield Simulators**, and **Battery Pack (S/P) Configuration**.
- **Electronics Design (EDA) Calculators:** Built advanced tools including **Filter Designers (RC/LC/Pi)**, **Resistor Divider** smart solvers, **Voltage/Current Margining (DAC)**, **Thermistor R-T Curves**, **IPC-2221 Trace Width**, **AWG Wire Ampacity**, **E-Series Finders**, and **Magnetics & Harness** drop calculators.
- **Hardware Interfacing: Web Serial Interface** providing real-time hardware stream monitoring and plotting directly in the browser.

## Education

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**Lovely Professional University (LPU)**

*Master of Technology (M.Tech) in Electrical Engineering (Power & Renewable Energy)*

**2022**

*Grade: A*

**University of Kashmir**

*Bachelor of Technology (B.Tech) in Electrical Engineering*

**2019**

*Grade: A*

## Selected Publications

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- *Ultracapacitor Integration with Regenerative Braking Systems in Electric Drive Trains* (ScienceDirect/Elsevier, 2023)
- *Experimental Implementation of Optimally Tuned Multifunctional RLMLS Driven VSc...* (Springer Link, 2023)
- *Application of V2V Energy Sharing in Electric Vehicles with Source Switching* (IEEE Xplore, 2022)
- *A New Modular Multilevel Converter Topology Using Flying Ultra-capacitor...* (Springer, 2021)